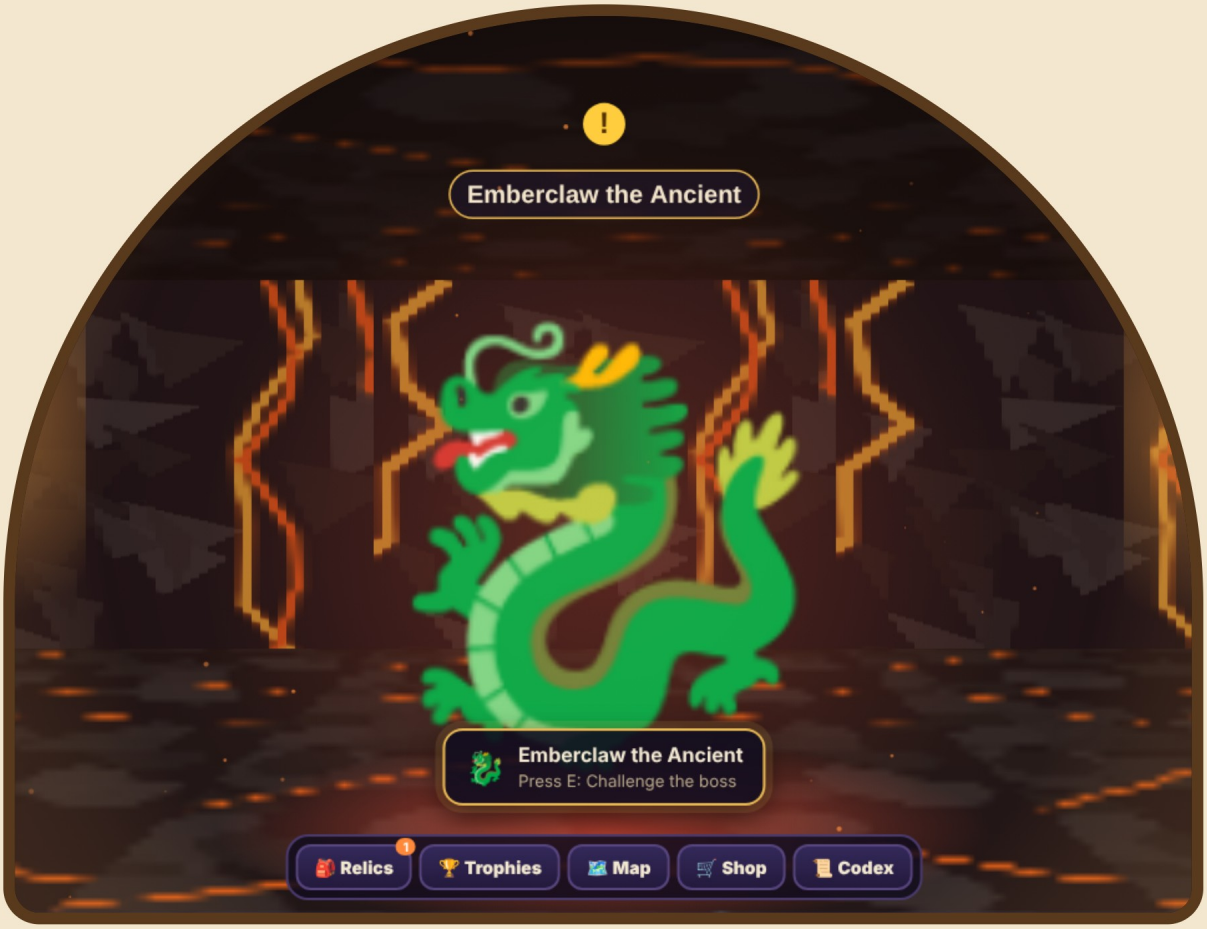
THE ADVENTURER'S HANDBOOK

EMBERDEEP

THE RIDDLE VAULTS



FIVE FLOORS. TWENTY GUARDIANS.
ONE VERY GRUMPY DRAGON.

MATHS · LOGIC · SCIENCE · WORDS & WORLD

This handbook belongs to _____

Brave thinker, clever spark, future Legend of Emberdeep

CHAPTER I

Your Quest

Deep beneath the hills lies **Emberdeep**, a dungeon built by riddlers. Its treasures are not guarded by swords. They are guarded by **questions**.

1 Explore

Wander the twisting tunnels by torchlight. Scoop up gems, read glowing rune stones, and find riddle chests.

2 Win four sigils

Every floor has four guardians. Take each one's trial: answer at least 3 of their 4 questions to win a golden sigil.

3 Unseal the door

With all four sigils, the sealed door at the far corner of the floor grinds open.

4 Defeat the boss

Behind the door waits the floor boss. Out-think it in a battle of brains.

5 Descend

Beat the boss and the stairs appear. Down you go, to a new floor and a new kind of puzzle.

A guardian waiting in the Mosslight Crypt. The yellow ! means you haven't won its sigil yet.



? **Lost?** Follow the **compass arrow** at the top of the screen. It always points to your next goal. Press **M** for the full map.

CHAPTER II

How to Move

On a keyboard

W A S D	Walk forward and back, and step sideways
← → ↑ ↓	Turn left and right, walk forward and back
Drag	Hold the mouse and drag to look around
E	Talk, open a chest, read a stone
Space	Jump
M	Open the map
1 2 3 4	Pick an answer
Enter	Check your answer, then go on
Esc	Close a window

On a tablet

	The round **joystick** (bottom left) walks and turns
USE	Talk, open, read
JUMP	Jump
Swipe	Drag the dungeon to look around, or tap a guardian to talk

Tip: answer with your finger on the big buttons, or type numbers on the keypad.

Reading the screen



CHAPTER III

The Five Floors

Each floor tests a different kind of thinking. Tick off each guardian and boss as you beat them!

Floor 1 — Mosslight Crypt — MATHS

Old stones that love to count.

- [ ] Numbra the Counter
- [ ] Toadmaster Tally
- [ ] The Stone Abacus
- [ ] The Ghost of Googol
- [ ] **BOSS: The Troll of Ten Thousand**

Floor 2 — Cogwork Halls — LOGIC

Every gear turns for a reason.

- [ ] Brasswhistle
- [ ] Cipher Sprite
- [ ] The Tick-Tock Owl
- [ ] Locksmith Mole
- [ ] **BOSS: The Clockwork Sphinx**

Floor 3 — Glowcap Caverns — SCIENCE

Where the mushrooms light the way.

- [ ] The Sporeling Sage
- [ ] Professor Snailsworth
- [ ] Echo the Bat
- [ ] The Crystal Golem
- [ ] **BOSS: Queen Silkspinner**

Floor 4 — The Rune Library — WORDS & WORLD

A million words, and some of them bite.

- [ ] Inkpaw
- [ ] Count Syllable
- [ ] The Elf Scribe
- [ ] Polyglot the Parrot
- [ ] **BOSS: The Lexicon Lich**

Floor 5 — The Dragon's Vault — EVERYTHING

Everything you have learned, all at once.

- [ ] The Genie of Guesses
- [ ] Cinder the Hatchling
- [ ] Sir Cumference
- [ ] The Tide Oracle
- [ ] **BOSS: Emberclaw the Ancient**

And then? Beneath Emberclaw's vault the dungeon begins again at **Depth II**: the same five floors with new mazes and harder questions. How deep can you go?

CHAPTER IV

Trials & Battles

Guardian trials

Four questions. Get **3 or more** right to win the guardian's sigil and a **relic**.

You get **3 tries** per question. Get all 4 right on your first try for a **perfect trial** and a rarer relic.

Already won? Visit again to practise for XP and gems.

Riddle chests

One question from any subject, with **2 tries**. Inside: a pile of gems, and sometimes a relic.

Rune stones

Glowing stones carved with secrets. Read all 15 to fill your **Codex**.

Boss battles



The boss has **6 health**. You have **3 hearts**.

Right answer: the boss loses 1 health.

Fast and first try: a **critical hit**, 2 health!

Wrong answer: you lose a heart. A shield charm blocks one.

The final blow is always a combination lock.

Out of hearts? You retreat to catch your breath. You keep all your XP, and you can challenge the boss again straight away.

⚡ Speed bonus

A bar shrinks at the top of each question. Answer on your first try before it runs out for bonus XP.

Pick-an-answer: 12 sec · numbers: 25 sec · words: 45 sec · locks: 2 min

💡 Hints

Stuck? The hint button gives a clue or knocks out a wrong answer. It halves the XP for that question, unless Hoot the Owl is with you.

CHAPTER V

The Questions

Four kinds of brain-work. Each subject has its own **level from 1 to 6**. Get 3 first-try answers in a row and it levels up. Miss twice in a row and it eases back down.

MATHS

- A dragon has 6 sacks with 9 gems in each. It gives 14 away. How many gems are left?
- What is 10% of 240?
- How many zeros are there in one billion?

LOGIC

- What comes next? 1, 1, 2, 3, 5, 8, ☐
- In a race, you overtake the runner in 2nd place. What place are you in now?
- Decode: ESBHPO (every letter moved 1 forward)

SCIENCE

- Why do we see lightning before we hear thunder?
- On the Moon, you drop a hammer and a feather together. Which lands first?
- What is the largest organ of the human body?

WORDS & WORLD

- Unscramble the letters: N O G A R D
- In te reo Māori, what does “whētū” mean?
- Which spelling is correct: Wensday, Wednesday or Wednsday?

Four Ways to Answer

Number pad. Type the number and press Check. ± makes a negative number, and the dot makes a decimal.

Pick an answer. Tap one of four choices, or press 1, 2, 3 or 4.

Word answer. Type the word for anagrams and secret codes.

Combination lock. Read the clues, then turn the three dials. See Chapter VI!



Answers to the examples: 40 · 24 · 9 · 13 · 2nd · DRAGON · light travels faster · both together · your skin · DRAGON · star · Wednesday

CHAPTER VI

Cracking the Lock

Locks guard the best treasure, and every boss ends with one. All three digits are different. Each clue compares a wrong guess with the real code.

Worked example

A	6	8	2	One digit is correct and in the right place.
B	6	1	4	One digit is correct but in the wrong place.
C	2	0	6	Two digits are correct but in the wrong place.
D	7	3	8	Nothing is correct.
E	7	8	0	One digit is correct but in the wrong place.

1. Start with clue **D**: nothing is correct. So **7, 3 and 8** are not in the code anywhere. Cross them off every clue.
2. Clue **E** (7 8 0): 7 and 8 are gone, so **0 is in the code**, but not in the last place.
3. Clue **A** (6 8 2): one of 6 or 2 is right *and in the right place*. If it were 6, it would be first. But clue **B** also has 6 first and says its digit is in the *wrong* place. So 6 is out, and **2 goes last**.
4. Clue **C** (2 0 6): 0 is in the wrong place in the middle, and the last place is taken by 2. So **0 goes first**.
5. Clue **B** (6 1 4): one of 1 or 4 is in the code, in the wrong place. Only the middle is left. 1 is already in the middle in this clue, so it can't be 1. **4 goes in the middle**.

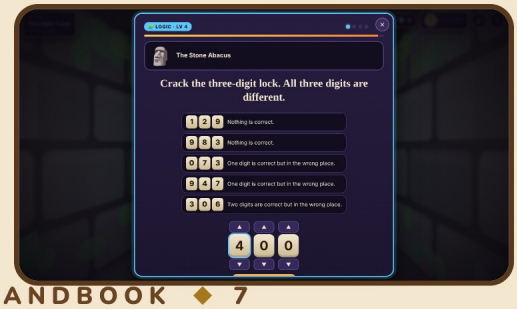
The code is 0 4 2

Detective tricks

- Always start with "nothing is correct".
- Write the code as three boxes and fill in what you know.
- A digit "in the wrong place" can't sit where that clue shows it.
- Check your answer against **every** clue before you turn the dials.

Your turn: practice lock

A	1	9	0	One digit is correct and in the right place.
B	4	1	0	Nothing is correct.
C	5	9	6	One digit is correct and in the right place.
D	2	0	5	One digit is correct but in the wrong place.
E	7	6	1	One digit is correct and in the right place.



CHAPTER VII

Rune-Codes

Guardians hide words in secret codes. Crack them with this toolkit.

The shift code

Every letter slides the same number of places along the alphabet. Julius Caesar used it 2,000 years ago!

Shift 1: **A→B, B→C … Z→A**. To decode, slide each letter **back**.

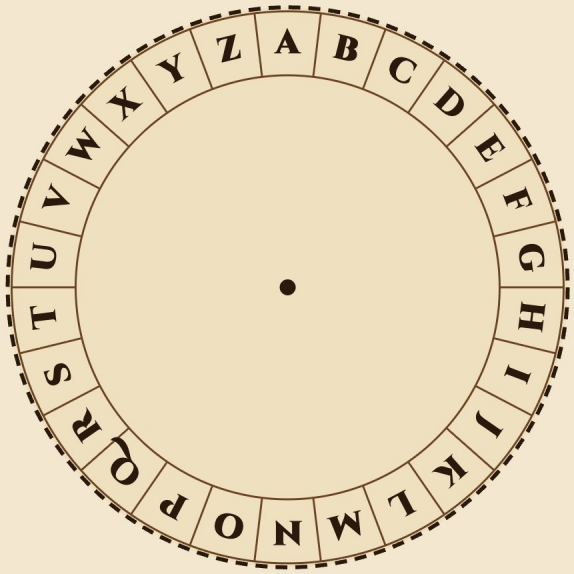
The mirror code

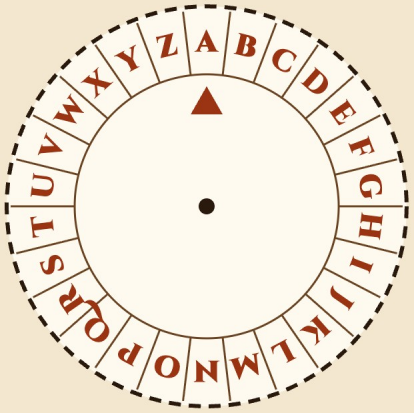
The alphabet folds in half: **A↔Z, B↔Y, C↔X** and so on. The same rule codes and decodes!

A	B	C	D	E	F	G	H	I	J	K	L	M
Z	Y	X	W	V	U	T	S	R	Q	P	O	N

Make a cipher wheel

Cut out both wheels along the dotted lines. Put the small wheel on top of the big one and push a split pin through both centres. Turn the small wheel so its arrow points to the big wheel's **A** plus the shift number, then read across.





Practice codes

SHIFT 1

ESBHPO

SHIFT 3

WRUFK

MIRROR CODE

PVB

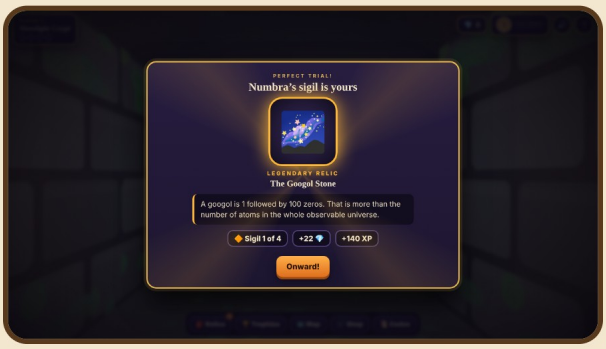
Answers: DRAGON · TORCH · KEY

CHAPTER VIII

Treasure & Trade

Relics

40 relics hide in Emberdeep, each with a true fact. Win them from guardians, chests and bosses. A duplicate turns into gems.



COMMON
16 to find · Most often

RARE
12 to find · About 1 in 4

EPIC
8 to find · About 1 in 10

LEGENDARY
4 to find · About 1 in 30

Bosses and **perfect trials** always give a rare relic or better.

💎 Gems

Pick up glittering gems as you explore. You also earn them from trials, chests, bosses, trophies (+10 each) and every new level (+15).

⭐ XP and Ranks

Every right answer earns XP. Your rank grows every two levels:

Torch Bearer → Riddle Rookie → Rune Reader → Vault Scout → Code Breaker → Sigil Seeker → Lore Keeper → Dungeon Sage → Master of Keys → Arch-Thinker → Legend of Emberdeep

The Goblin Market

Ember Lantern — Free A warm orange glow. You start with it.	**Frostfire & Witchlight Lanterns** — 60 each Cool blue or eerie green light.	**Royal Violet Lantern** — 90 A rich purple shimmer.
Prism Lantern — 200 Shifts slowly through every colour.	**Hoot the Owl** — 120 Hints no longer cost you any XP.	**Flick the Fox** — 120 Sniffs out gems from much further away.
Shellby the Turtle — 80 8 extra seconds for speed bonuses.	**Cinderpip the Dragonling** — 250 One extra heart in boss fights.	**Shield Charm** — 40 Blocks one wrong answer in a boss fight. Carry up to 3.

Prices are in gems. Companions follow you around; only one can come along at a time.

COLLECTION LOG

The Relic Vault

Tick each relic as you find it. Can you collect all 40?

- ☐  **Counting Beads** — COMMON
- ☐  **Everburn Candle** — COMMON
- ☐  **Lodestone Shard** — COMMON
- ☐  **Coin of Halves** — COMMON
- ☐  **Fossil Fang** — COMMON
- ☐  **Spiral Shell** — COMMON
- ☐  **Scroll of Palindromes** — COMMON
- ☐  **Echo Bell** — COMMON
- ☐  **Bubbling Flask** — COMMON
- ☐  **Glowcap** — COMMON
- ☐  **Moa Feather** — COMMON
- ☐  **True Compass** — COMMON
- ☐  **Die of Twenty** — COMMON
- ☐  **Sandglass** — COMMON
- ☐  **Skeleton Key** — COMMON
- ☐  **Silk Thread** — COMMON
- ☐  **Prime Orb** — RARE
- ☐  **Alchemist's Still** — RARE
- ☐  **Obsidian Blade** — RARE
- ☐  **Tuatara Scale** — RARE
- ☐  **Neverthaw Ice** — RARE
- ☐  **Horn of Fibonacci** — RARE
- ☐  **Mirror of Symmetry** — RARE
- ☐  **Hexagon Seal** — RARE
- ☐  **Meteor Chip** — RARE
- ☐  **Rosetta Chip** — RARE
- ☐  **Bottled Lightning** — RARE
- ☐  **Octave String** — RARE
- ☐  **Crown of a Million** — EPIC
- ☐  **Helix Charm** — EPIC
- ☐  **Ring of Saturn** — EPIC
- ☐  **Möbius Band** — EPIC
- ☐  **The Grain Board** — EPIC
- ☐  **Eye of Pi** — EPIC
- ☐  **Dragonscale** — EPIC
- ☐  **Farseer Lens** — EPIC
- ☐  **The Googol Stone** — LEGENDARY
- ☐  **Infinity Loop** — LEGENDARY
- ☐  **Pocket Black Hole** — LEGENDARY
- ☐  **Heart of Emberdeep** — LEGENDARY

Common · Rare · Epic · Legendary

COLLECTION LOG

The Trophy Hall

Thirty trophies, each worth 10 gems. Tick them off and write the date you earned them.

- ☐ 👣 **First Steps** — Answer your first question correctly — _____
- ☐ 🏀 **Unstoppable** — Get 10 first-try answers in a row — _____
- ☐ 💯 **Century** — Answer 100 questions correctly — _____
- ☐ ⚡ **Quick Thinker** — Earn 10 speed bonuses — _____
- ☐ 🕵️ **Code Breaker** — Decode 5 secret rune-codes — _____
- ☐ 📖 **Lore Seeker** — Read 5 rune stones — _____
- ☐ 🏅 **Boss Buster** — Defeat your first floor boss — _____
- ☐ ⬇️ **Going Down** — Reach the third floor — _____
- ☐ 🎒 **Collector** — Find 10 different relics — _____
- ☐ 👑 **Completionist** — Find all 40 relics — _____
- ☐ 🌈 **Legendary Find** — Find a legendary relic — _____
- ☐ 🎓 **Master of Maths** — Reach maths level 5 — _____
- ☐ 🔬 **Master of Science** — Reach science level 5 — _____
- ☐ 🦘 **Bouncy** — Jump 50 times — _____
- ☐ 🗺️ **Cartographer** — Explore 90% of a floor — _____
- ☐ 🔥 **On Fire** — Get 5 first-try answers in a row — _____
- ☐ 🌟 **Legendary Streak** — Get 20 first-try answers in a row — _____
- ☐ 🧠 **Big Brain** — Answer 500 questions correctly — _____
- ☐ 🔐 **Locksmith** — Crack 5 combination locks — _____
- ☐ 🧰 **Treasure Hunter** — Open 10 treasure chests — _____
- ☐ 📚 **Loremaster** — Read every rune stone — _____
- ☐ 🛡️ **Flawless** — Defeat a boss without losing a heart — _____
- ☐ 🐲 **Dragon Tamer** — Defeat Emberclaw on the fifth floor — _____
- ☐ 🏛️ **Curator** — Find 25 different relics — _____
- ☐ 💜 **Epic Find** — Find an epic relic — _____
- ☐ 💰 **Gem Hoarder** — Collect 500 gems in total — _____
- ☐ 🧩 **Master of Logic** — Reach logic level 5 — _____
- ☐ 🔪 **Master of Words** — Reach words level 5 — _____
- ☐ 🐾 **Best Friend** — Adopt a companion — _____
- ☐ 🎖️ **Seasoned** — Reach adventurer level 10 — _____

DEEPEST FLOOR	BEST STREAK	HIGHEST LEVEL
_____	_____	_____

FOR GROWN-UPS

Parents & Teachers

Emberdeep is a single web page that adapts to the player, saves on the device, and sends nothing anywhere.

How the difficulty adapts

Maths, logic, science and words each keep their own level from 1 to 6. A subject moves up after three first-try answers in a row and eases down after two misses in a row, so every player settles where they are challenged but not stuck. After the fifth floor the whole dungeon repeats one notch harder. The year levels below are rough guides to the maths, not curriculum mappings.

1. Around Year 4: times tables, adding and halving
2. Around Year 5, where everyone starts: place value, fractions of amounts, Roman numerals
3. Year 5 to 6: order of operations, area, negative numbers, the -illions
4. Year 6 to 7: percentages, decimals, primes and factors, powers
5. Around Year 7: adding fractions, ratio, volume, simple equations
6. Around Year 8: binary, probability, square roots, two-step algebra

Checking progress

Open **Codex › Progress** to see each subject's level and first-try accuracy, total correct answers, best streak and deepest floor.

To start over, choose **Start a new adventure** on that screen and tap it twice.

Saving and privacy

Progress saves automatically in this browser on this device. There are no accounts, ads or tracking, and the game makes no network requests. Clearing browser data or using a private window starts fresh.

Playing together

- Ask how they got an answer, even a right one.
- Wrong answers show a worked explanation: read it together.
- Keep paper and a pencil nearby for locks and codes.
- Relic facts and rune stones make good dinner-table questions.

Play online at **joelal447.github.io/emberdeep**, or open the game file in any web browser.

Made as a family project, with help from Claude.